

Machine Learning and the Physical World

Lecture 9 : Reinforcement Learning

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9th of November, 2026

http://carlhenrik.com

A for grades 85% and above

B for grades between 75% and 84.99%

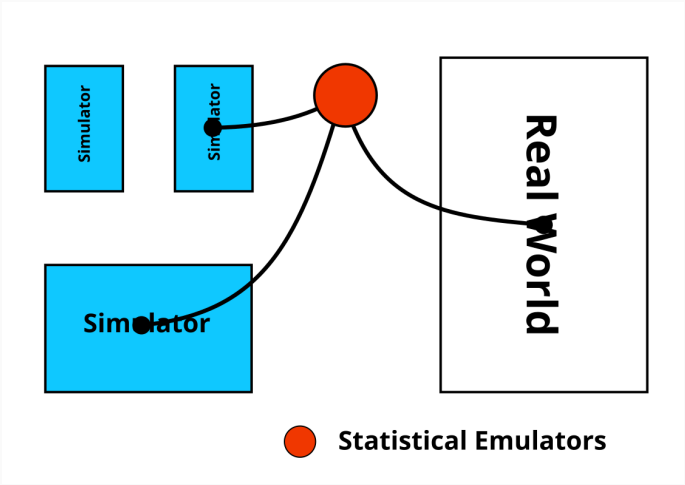
C for grades between 65% - 74.99%

D for grades between 50% - 64.99%

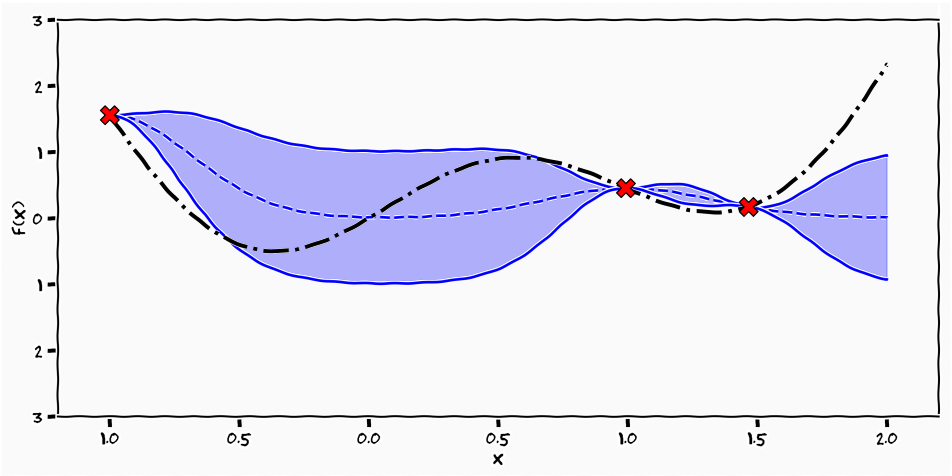
E for grades between 0% - 49.99%

Introduction

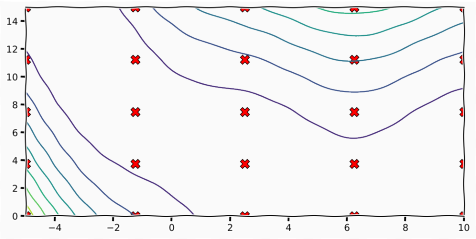
Simulation and Emulation



Emulators



Branin Function

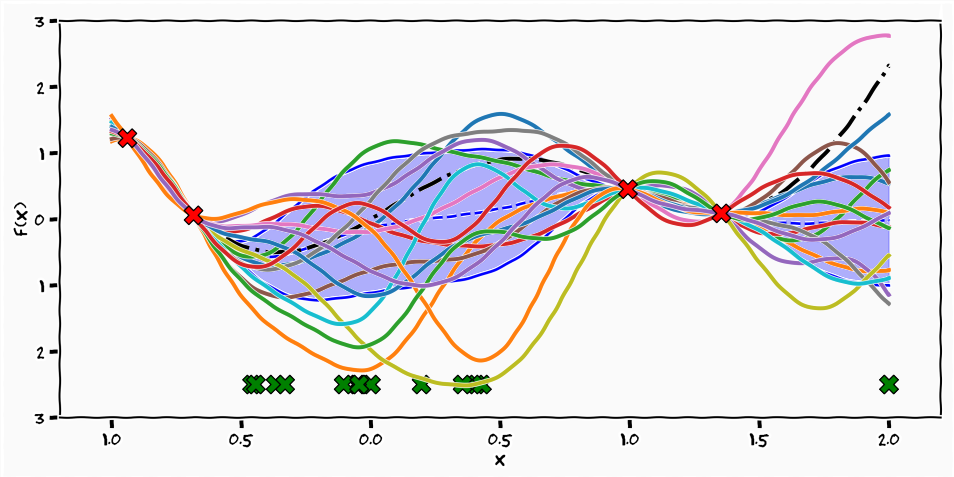


Fit a GP surrogate

$$k(\mathbf{x}_i, \mathbf{x}_j) = k_{\text{Exp}}(\mathbf{x}_i, \mathbf{x}_j) + k_{\text{White}}(\mathbf{x}_i, \mathbf{x}_j) + k_{Bias}$$

$$\mathbb{E}[f(U)] \approx \mathbb{E}[p(\mathbf{f} \mid \mathbf{Y}, \mathbf{X}, \mathbf{X}_{\text{samples}}) = 70.25$$

Surrogate Uncertainty



Data Acquisition

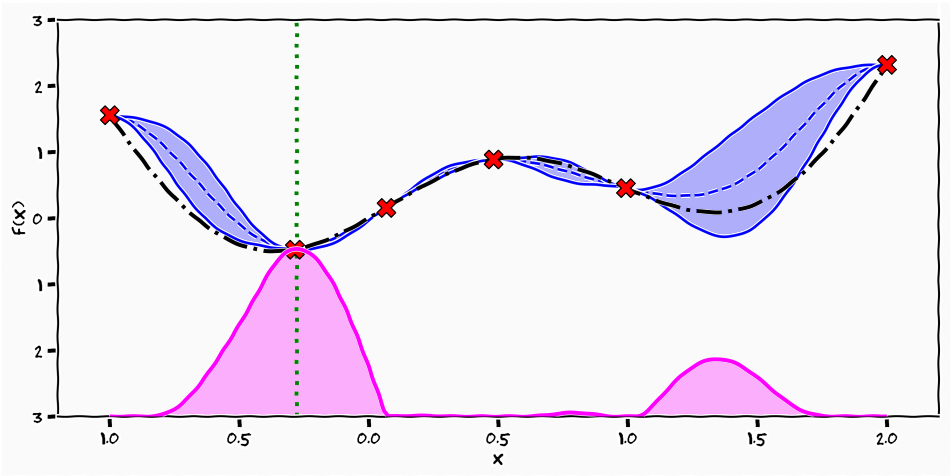
- Define a function that defines the utility of observing each location

$$u(x, f(x^{(*)}), \mathcal{M}_n)$$

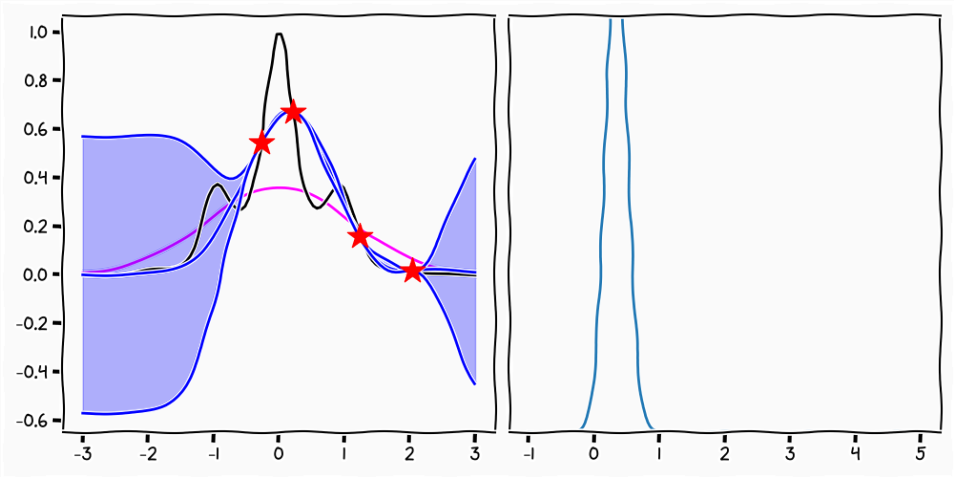
- Define the acquisition function as the expected utility

$$\begin{aligned}\alpha(x; \{x_i, y_i\}_{i=1}^n, \mathcal{M}_n) &= \mathbb{E}_{p(f)}[u(x)] \\ &= \int u(x, f(x^{(*)}), \mathcal{M}_n) p(f \mid \{x_i, y_i\}_{i=1}^n) \mathrm{d}f\end{aligned}$$

Optimisation



Quadrature



Today

November 9, 2026

Reinforcement Learning

Supervised Learning predict output from input

$$\mathcal{D} = \{y_i, x_i\}_{i=1}^{N}$$

$$p(\mathbf{y}|\mathbf{x}) = \int p(\mathbf{y}|\mathbf{x}, \theta) p(\theta) \mathrm{d}\theta$$

Supervised Learning predict output from input

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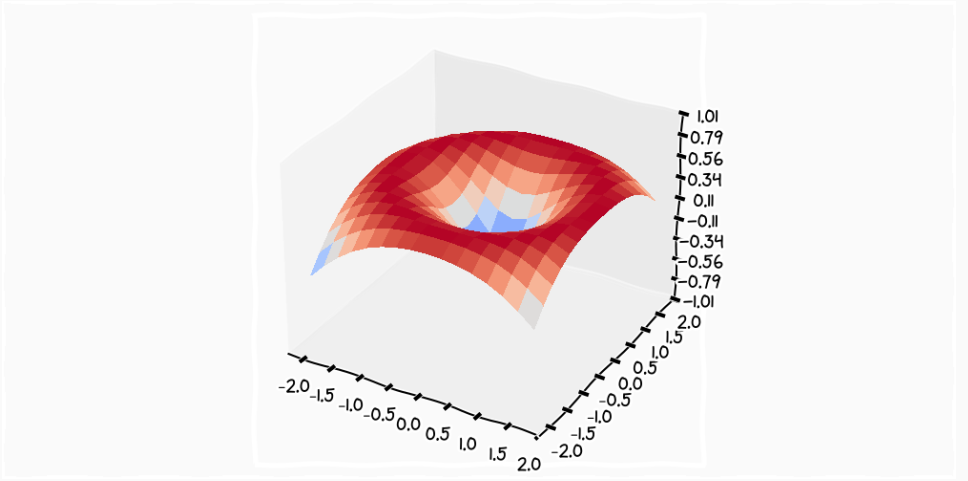
$$p(\mathbf{y}|\mathbf{x}) = \int p(\mathbf{y}|\mathbf{x}, \theta) p(\theta) \mathrm{d}\theta$$

Unsupervised Learning model the data

$$\mathcal{D} = \{y_i\}_{i=1}^N$$

$$p(\mathbf{y}) = \int p(\mathbf{y}|\mathbf{x}, \theta) p(\mathbf{x}) p(\theta) \mathrm{d}\mathbf{x} \mathrm{d}\theta$$

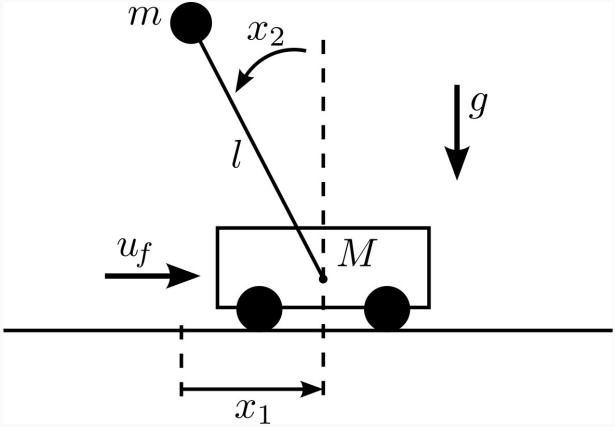
Structure


1.01
0.79
0.56
0.34
0.11
−0.11
−0.34
−0.56
−0.79
−1.01
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
−2.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0

Reinforcement Training



Inverted Pendlum



Reinforcement Learning





Can we learn without specifying how the task should be achieved by providing, rewards (positive) and punishment (negative)?

Formalism

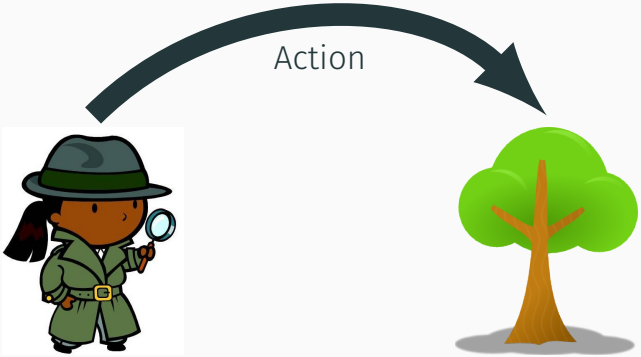


Formalism

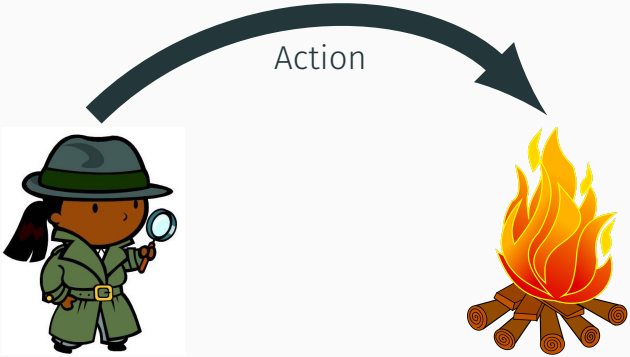




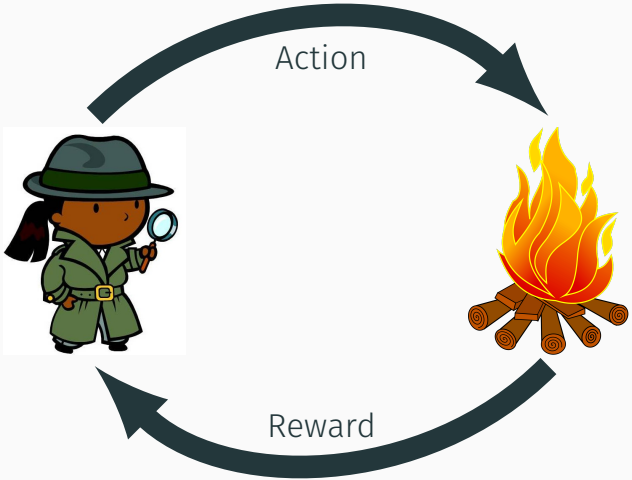
Formalism



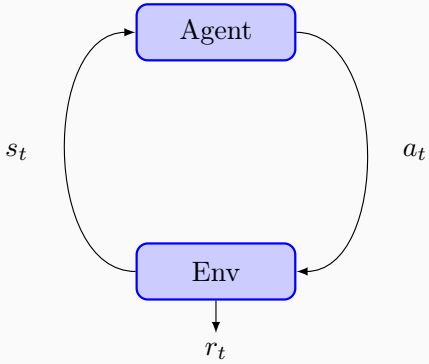
Formalism



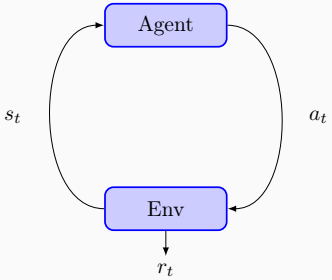
Formalism



The Agent and the Environment



Reinforcement Learning



$$\text{argmax}_{a_0, a_1, \ldots} \sum_{t=0}^{\infty} r_t$$

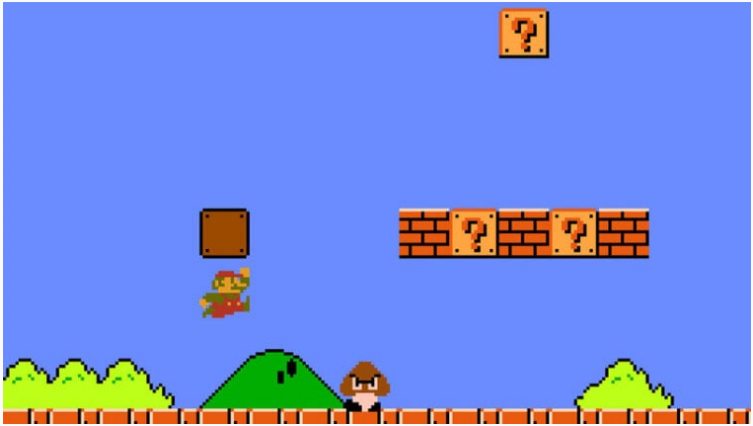
- the agent takes actions in the environment
- actions change the state of the environment
- the state produces a reward

How is this different?

$$p\left(\sum_{t=0}^{T} r_t \mid s_0, a_0, a_1, \ldots, a_t\right)$$

Lets just collect lots and lots of data and turn it into a regression problem

Super Mario



When to do RL?

- When the number of possible sequences of input is very large

When to do RL?

- When the number of possible sequences of input is very large
- When acquiring data is hard and expensive

When to do RL?

- When the number of possible sequences of input is very large
- When acquiring data is hard and expensive
- When its hard to define the ”structure” of the problem explicitly as a cost

Markov Decision Processes

Definition (Markov Decision Process)

A Markov Decision Process is a 4-tuple (S, A, T, R)

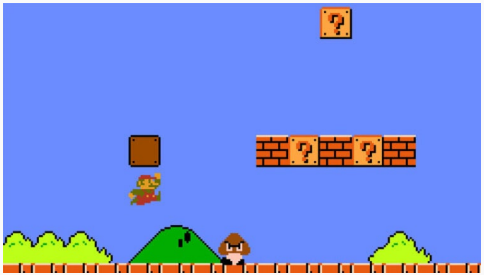
- S is the State space
- A is the Action space
- T the state transition function
- R the reward function

Markov Decision Process

$$T(s_t, a_t) = p(s_{t+1} \mid s_t, a_t)$$

- RL provides solutions to Markov Decision Processes
- We must define our state space in such a way that the environment forms an MDP

State Space



State Space

- position and velocity of Mario
- the score
- number of coins collected
- time remaining
- Goomba position & velocity, state
- ....

Action Space



Action Space

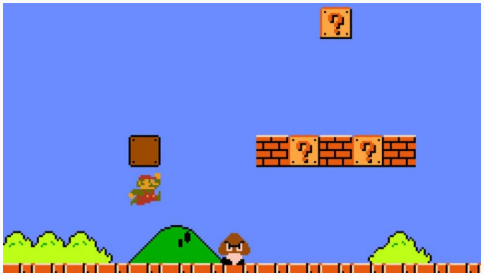
- The controller has

$$a, b, \uparrow, \downarrow, \leftarrow, \rightarrow$$

- Can press certain buttons together

$$A = \underbrace{\{0, 1, \ldots, 8\}}_{\text{directions}} \times \underbrace{\{0, 1, 2, 3\}}_{\text{buttons}}$$

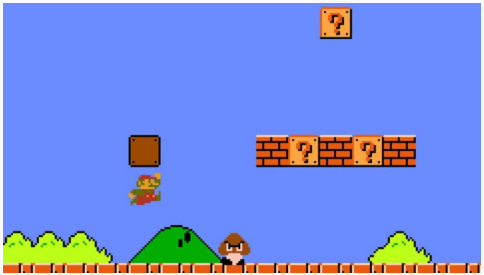
Transition Function



Transition Function

$$T : S \times A \to \Delta S$$

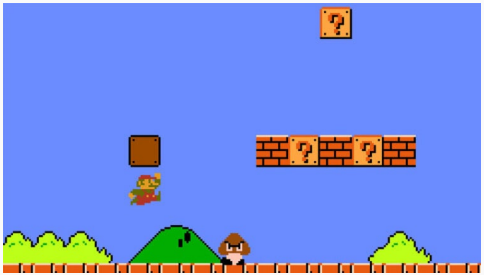
Reward Function



Reward Function

- beating the level $(0, 1)$

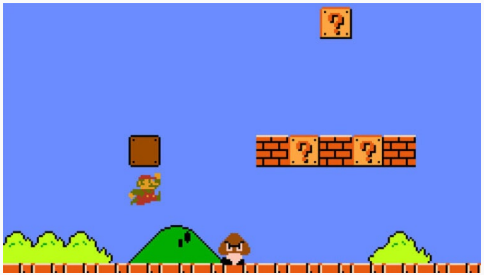
Reward Function



Reward Function

- beating the level $(0, 1)$
- total score

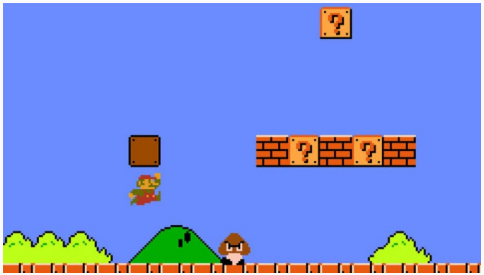
Reward Function



Reward Function

- beating the level $(0, 1)$
- total score
- beating the level $(0, 1) + 0.01 \times \text{score}$

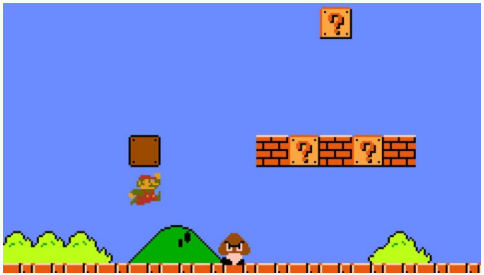
Reward Function



Reward Function

- beating the level $(0, 1)$
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- beating the level $(0, 1) + 0.01 \times$ score
- beating the level/time

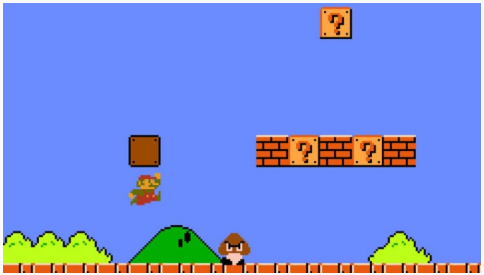
Reward Function



Reward Function

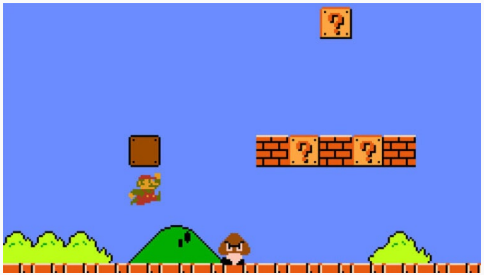
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- ....

Markov Decision Processes



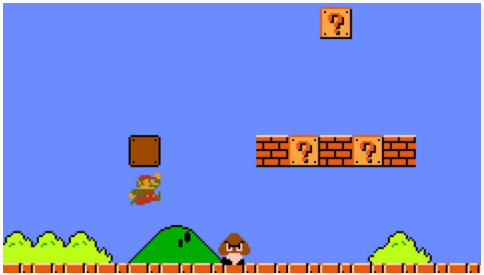
- think carefully about constructing the environment

Markov Decision Processes



- think carefully about constructing the environment
- think carefully about how to structure reward signal

Markov Decision Processes



- think carefully about constructing the environment
- think carefully about how to structure reward signal
- now we have a problem to approach with RL

Reinforcement Learning

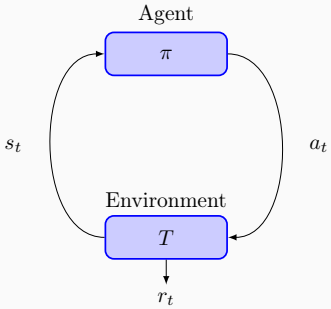
$$G = \sum_{t=0}^{\infty} r_t$$

- Task of RL is to design an agent to maximise return G

$$G = \sum_{t=0}^{\infty} \gamma^t r_t$$

- Task of RL is to design an agent to maximise return G
- We generally have a preference of cookies now compared to cookies later which we introduce by $\gamma \in (0, 1]$ called the discount factor

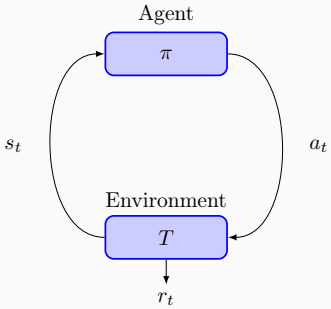
Agent



- The agent acts according to a policy π

$$\pi : S \to A$$

Agent



- The agent acts according to a policy π

$$\pi : S \to A$$

- Goal is to learn the optimal policy

$$\pi_* = \max_\pi \sum_{t=0}^{\infty} \gamma^t r_t$$

The Learning Problem

- We do not know the dynamics of the environment

$$p(s_{t+1} \mid s_t, a_t)$$

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$$p(a_t \mid s_t)$$

- We not might even know the reward

$$p(r_t \mid s_t)$$

The Learning Problem

- We do not know the dynamics of the environment

$$p(s_{t+1} \mid s_t, a_t)$$

- We do not know the policy

$$p(a_t \mid s_t)$$

- We not might even know the reward

$$p(r_t \mid s_t)$$

- We have no data to learn from

What do we want to learn?

- we want to interact with the environment as little as possible

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What do we want to learn?

- we want to interact with the environment as little as possible
- we only want to know the dynamics up to a level where the policy will act
- we only need to understand how to act to get a high reward
- learning the dynamic and policy are just proxies

1. initially we do not have any data

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2. we usually start off with an initial experiment

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3. when we get data we can update our dynamics and policy

1. initially we do not have any data
2. we usually start off with an initial experiment
3. when we get data we can update our dynamics and policy
4. how to acquire data from the knowledge that we have?

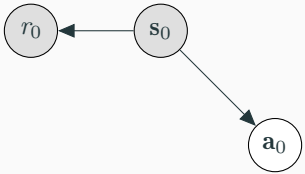
Model Based Reinforcement Learning

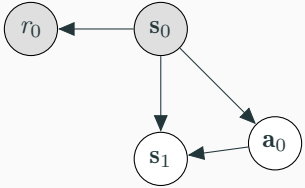
$$p(s_{t+1} \mid s_t, a_t)$$
$$p(a_t \mid s_t)$$
$$p(r_t \mid s_t)$$

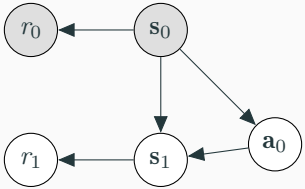
- Model the dynamics and the policy directly
- All the benefits of having uncertainty over each specific component
  - bias with prior knowledge
  - efficient exploration
- Just normal statistical inference :-)

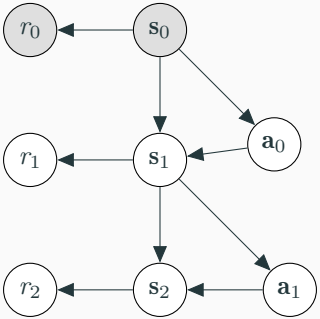

$\mathbf{s}_0$


r_0
$\mathbf{s}_0$

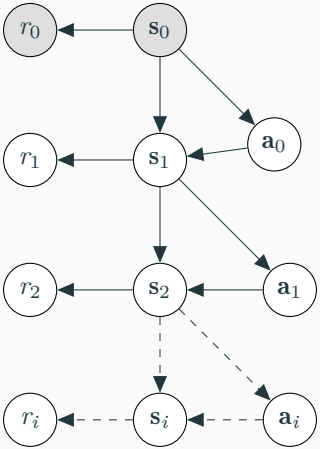

r_0
$\mathbf{s}_0$
$\mathbf{a}_0$


r_0
$\mathbf{s}_0$
$\mathbf{a}_0$
$\mathbf{s}_1$

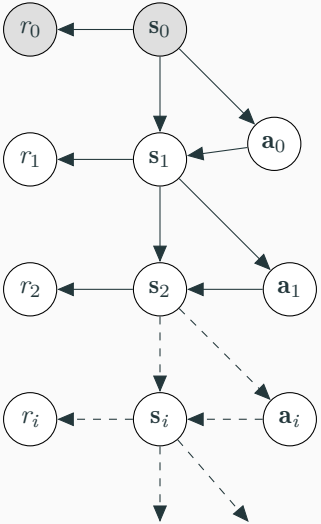

r_0
$\mathbf{s}_0$
$\mathbf{a}_0$
r_1
$\mathbf{s}_1$


r_0
$\mathbf{s}_0$
$\mathbf{a}_0$
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$\mathbf{a}_1$

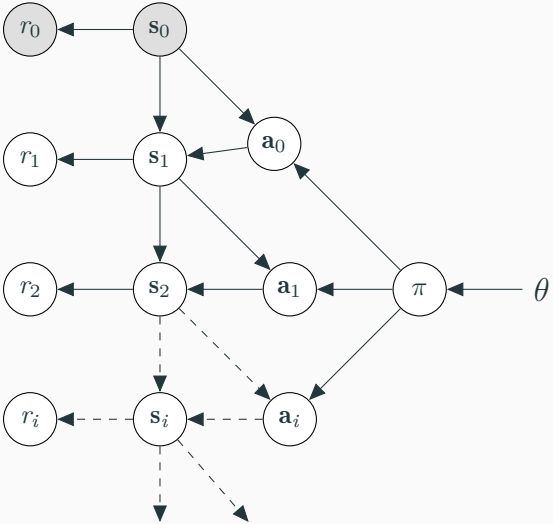
Reinforcement Learning



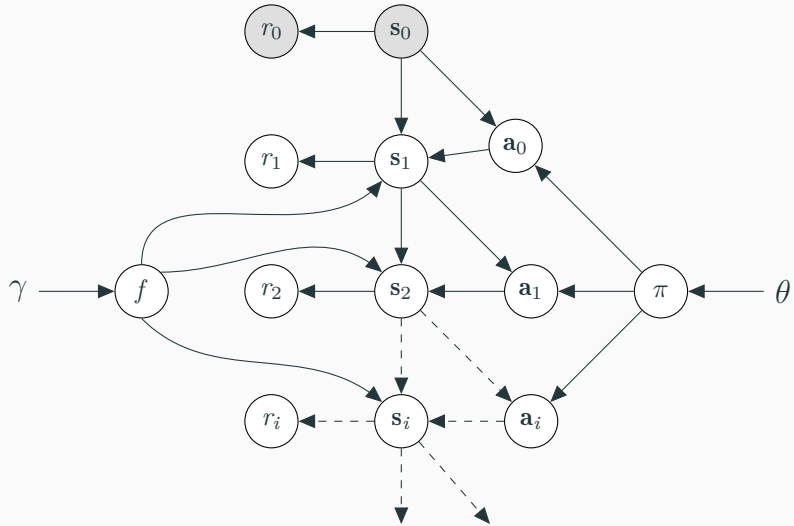
Reinforcement Learning



Reinforcement Learning



Reinforcement Learning



$$
\begin{aligned}
p(\mathbf{s}_{1,\ldots,T}, a_{0,\ldots,T-1}, r_{0,\ldots,T}, f, \pi, \gamma, \theta | \mathbf{s}_0) = \\
p(\mathbf{s}_T | a_{T-1}, s_{T-1}, f) p(a_{T-1} | s_{T-1}, \pi) \ldots \\
\vdots \\
p(\mathbf{s}_2 | a_1, s_1, f) p(a_1 | s_1, \pi) \\
p(\mathbf{s}_1 | a_0, s_0, f) p(a_0 | s_0, \pi) \\
p(f | \gamma) p(\pi | \theta) p(\gamma) p(\theta)
\end{aligned}
$$

- if we want to learn dynamics and policy we need to marginalise out f, π, θ and γ
- very very hard problem as uncertainty propagated a long way

$$\mathbb{E}[r_t] = \int \int p(r_t \mid s_{t+1}) p(s_t + 1 \mid s_t, a_t) p(a_t \mid s_t) \mathrm{d}a_t \mathrm{d}s_{t+1}$$

$$\pi_* = \max_\pi \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t r_t \mid a_t \sim p(a_t \mid s_t)\right]$$

- We want to avoid explicitly learning a policy and a reward model
- Directly maximise the expected reward

Q-Learning

- Define value function V,

$$V_\pi(s_0) = \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t r_t \mid a_t \sim \pi(s_t)\right]$$

Q-Learning

- Define value function V,

$$V_\pi(s_0) = \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t r_t \mid a_t \sim \pi(s_t)\right]$$

- Lets move out the first action and define the function,

$$\tilde{V}_\pi(s_0, a_0) = \mathbb{E}[r_0 \mid a_0] + \mathbb{E}\left[\sum_{t=1}^{\infty} \gamma^t r_t \mid a_t \sim \pi(s_t)\right]$$

Q-Learning

- Define value function V,

$$V_\pi(s_0) = \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t r_t \mid a_t \sim \pi(s_t)\right]$$

- Lets move out the first action and define the function,

$$\tilde{V}_\pi(s_0, a_0) = \mathbb{E}[r_0 \mid a_0] + \mathbb{E}\left[\sum_{t=1}^{\infty} \gamma^t r_t \mid a_t \sim \pi(s_t)\right]$$

- This we will refer to as the Q-function,

$$Q_\pi(s_0, a_0) = \tilde{V}_\pi(s_0, a_0)$$

Q-Learning

- How should we pick our policy?

Q-Learning

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- There are many things where RL is a suitable approach

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References



Sutton, Richard S. and Andrew G. Barto (1998). ***Introduction to Reinforcement Learning***. 1st. Cambridge, MA, USA: MIT Press.